Semester project for course: Pattern Recognition
Sem: Monsoon 2026
Deadline: 5, November, 2026

Instructions

1. The project topic must be selected based on a real-world problem
2. Project team formation - google sheet - is available here. You can also add your project topic in the same sheet.
3. You may use any dataset (from the list or elsewhere) provided it satisfies the following conditions
    a. No. of attributes >= 10
    b. No. of instances >= 1000
    c. Problem - Multi-Category Classification only (at least 3 classes) (Supervised Learning)
4. Repositories of datasets **(Suggestions only)**
    a. EMAHA-DB (IIITS - HMA Lab) or search in Kaggle for EMAHA DB
    b. NINAPRO Datasets
    c. UCI machine learning Repository
    d. Letter Recognition - UCI Machine Learning Repository
    e. Kaggle - eg. Kaggle CS2 telemetry data:
    f. MNIST Dataset
    g. Fashion MNIST
    h. CIFAR-10 datasets
    i. hyperspectral dataset | IEEE DataPort
    j. WHU-HSI: https://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

5. The main content of the project should be focused on the pattern recognition aspects of the problem
   Specifically an improved classification performance through
    - **Feature Extraction**
      or
    - **modification/innovations in the existing classical pattern recognition methods**
    - **Deep learning methods can be used for comparison only**

   Should be demonstrated through valid comparison studies.
6. **The individual contributions should be made very clear and must be presented**
7. The final report should be **submitted as a team** and the **individual contributions should be clearly articulated** in this report in the appendix section.
8. In the report, **the matlab/python code should be provided as a repository link (should be active until grades are awarded)**
9. For the evaluation purpose, **the code should be shared and a file named Main.m should be included which if executed should run the whole program without any need of intervention and it should display all the desired results.**
10. **For reporting results:**

A. Use a stated train/validation/test split or k-fold cross-validation, stratified for imbalanced classes.
B. Prevent data leakage: fit scaling, feature selection, and PCA on training data only.
C. Report accuracy plus macro-F1, a confusion matrix, and per-class precision/recall.
D. Report mean ± standard deviation over multiple runs or folds.

11. Project Grades are based on the following deliverables: (see course plan)
    a. **Executable code submitted on time: 60% (max)**
    b. **Final report : 30% (max)**
    c. **Final presentation: 10% (max)**
12. The individual scores may vary depending on performance and contributions.
13. **No two teams can have the same topic**
14. **Any instances of plagiarism or copying codes/ideas from other teams may lead to 0 final score.**